

1.

a. This is a cross-sectional cohort study, which is an observational study. The exposure variable is smoking status, and the disease variable is liver cancer status.

b.

Relative risk is the risk in the exposed group over the risk in the unexposed group. We find the risk by looking at the proportion of smokers with liver cancer (30%), and the proportions of smokers without liver cancer (10%). This gives us:

$$\frac{\text{Risk for exposed group}}{\text{Risk for unexposed group}} = \frac{\frac{90}{300}}{\frac{70}{700}} = \frac{0.3}{0.1} = 3$$

Smokers have 3 times the chance of developing liver cancer as opposed to non-smokers.

We now calculate the odds ratio, which is the odds of liver cancer in smokers vs. the odds of liver cancer in non-smokers:

$$\text{Odds Ratio} = \frac{\frac{90}{210}}{\frac{70}{630}} = \frac{0.428}{0.111} = 3.86$$

The odds of having liver cancer are 3.86 higher for smokers than they are for non smokers.

c.

	Having liver cancer	Not having liver cancer	Total
Smoker	90	210	300
Non-smoker	70	630	700
Total	160	840	1000

First we find the standard error of the odds ratio:

$$SE = \sqrt{\frac{1}{90} + \frac{1}{210} + \frac{1}{70} + \frac{1}{630}} = 0.178$$

Now for the 95% CI for the log odds ratio

$$\log(3.86) \pm 1.96 * 0.178$$

$$(1.00, 1.70)$$

It appears that the relationship between smoking and liver cancer is significant in this population, as our 95% confidence interval does not pass through zero.

2.

a.

This is another cross-sectional cohort study. The exposure variable is smoking status, and the disease variable is lung cancer status.

b.

	Having lung cancer	Not having lung cancer	Total
Smoker	80	120	200
Non-smoker	40	760	800
Total	120	880	1000

We calculate the odds ratio:

$$OR = \frac{\frac{80}{120}}{\frac{40}{760}} = \frac{0.667}{0.053} = 12.6$$

The odds of smokers in this sample having lung cancer is 12.6 times higher for the smoking group than the non-smoking group.

We find the standard error next:

$$SE = \sqrt{\frac{1}{80} + \frac{1}{120} + \frac{1}{40} + \frac{1}{760}} = 0.217$$

So, the 95% CI for the log odds ratio is:

$$\log(12.6) \pm 1.96 * 0.217 = 2.53 \pm 0.425 = (2.10, 2.96)$$

The relationship between smoking and lung cancer appears to be significant as our 95% confidence interval does not pass through zero.

3.

To perform stratified randomization, you form strata using various factors (age, gender, race, etc.) then use permuted block randomization in each strata. For example, if you stratify by gender, you then perform permuted block randomization on the Male group, then the Female group. The advantage of stratified randomization is making the treatment groups similar, and allowing better estimates of treatment difference due to the strata. The disadvantage is that if you have too many prognostic factors, you could end up with too few patients in some strata, which would be a weak design.

4.

a.

We will determine the sample size as follows:

$$z_{\frac{\alpha}{2}} = 1.96 \text{ for two-sided}$$

$$z_{\beta} = 0.84 \text{ for 80\% power}$$

$$\sigma_Y = 5$$

$$\Delta = 2$$

$$n = \left(\frac{(Z_{\alpha/2} + Z_{\beta})^2 * \sigma_Y^2 * 4}{\Delta^2} \right)$$

$$n = \frac{(1.96 + 0.84)^2 * 5^2 * 4}{2^2} = \frac{7.84 * 25 * 4}{4} = 196$$

b.

We will determine the power if 100 patients are used as follows. First, we rearrange the formula solving for power.

$$n = \left(\frac{(Z_{\alpha/2} + Z_{\beta})^2 * \sigma_Y^2 * 4}{\Delta^2} \right) \text{ and } P(T \geq z_{\alpha} | \Delta = \Delta_A) = 1 - \beta \text{ for determining power.}$$

$$\frac{\Delta_A}{\sigma_Y} * \sqrt{\frac{2}{n}} = Z_{\alpha/2} + Z_{\beta}$$

We now solve for Z_{β}

$$\frac{2}{5} * \sqrt{\frac{2}{100}} = 1.96 + Z_{\beta}$$

$$Z_{\beta} = 0.05656 - 1.96 = -1.903$$

Using R, we determine β with the critical value -1.903:

```
> pnorm(-1.903)
[1] 0.02852027
```

So, our power is $1 - 0.029 = 0.971$.

With 100 patients in each group, the power is 97.1%

5.

a.

First, we construct a table of the expected counts:

$$E_{yes} = \frac{500 * 500}{1500} = 166.67$$

$$E_{no} = \frac{1000 * 500}{1500} = 333.33$$

Response	Treatment 1	Treatment 2	Treatment 3	Total
yes	166.67	166.67	166.67	500
no	333.33	333.33	333.33	1000
Total	500	500	500	1500

The chi-square test statistic formula is:

$$X^2 = \frac{\sum_{cells} (O_j - E_j)^2}{E_j}$$

$$X^2 = \frac{(150 - 166.67)^2}{166.67} + \frac{(350 - 333.33)^2}{333.33} + \frac{(125 - 166.67)^2}{166.67} + \frac{(375 - 333.33)^2}{333.33} + \frac{(225 - 166.67)^2}{166.67} + \frac{(275 - 333.33)^2}{333.33}$$

$$X^2 = 1.67 + 0.83 + 10.42 + 5.21 + 20.41 + 10.21 = 48.75$$

Our degrees of freedom is 2, noted by $(2\text{rows} - 1)(3\text{columns} - 1) = 1 * 2$

So, we obtain our critical value from a chi-square distribution table, or we can use R:

```
> 1 - pchisq(48.75, 2)
[1] 2.594613e-11
```

This is less than our level of 0.05 per the instructions, so we determine that H_A is not all the same as H_0 .

b.

We apply Bonferroni correction to accomplish this problem. We calculate our response rates for the 3 treatments:

$$p_1 = \frac{150}{500} = 0.3$$

$$p_2 = \frac{125}{500} = 0.25$$

$$p_3 = \frac{225}{500} = 0.45$$

We now use Bonferroni correction for 3 pairwise comparisons, so our alpha is:

$$\alpha = \frac{0.05}{2} = 0.025 \text{ and corresponding critical value is } z_{\alpha/2} = 2.24$$

```
> qnorm(1 - 0.025/2)
[1] 2.241403
```

So, we are interested if $T_{1j} \geq 2.24$.

$$T_{1j} = 2 \frac{\sqrt{n_1 * n_j}}{n_1 + n_j} (\sin^{-1} \sqrt{p_j} - \sin^{-1} \sqrt{p_1})$$

Comparing treatment 1 and 2:

$$T_{12} = 2 \sqrt{\frac{500 * 500}{500 + 500}} (\sin^{-1} \sqrt{0.25} - \sin^{-1} \sqrt{0.30}) = 2 * 15.81 (0.580 - 0.524) = 1.771$$

1.771 is less than 2.24, so the treatments are not significantly different. We now compare treatment 1 and 3:

$$T_{13} = 2 \sqrt{\frac{500 * 500}{500 + 500}} (\sin^{-1} \sqrt{0.45} - \sin^{-1} \sqrt{0.30}) = 2 * 15.81 (0.735 - 0.580) = 1.771 = 4.901.$$

4.901 is greater than 2.24, so treatments 1 and 3 are significantly different from one another, with treatment 3 being a more effective choice.

6.

a.